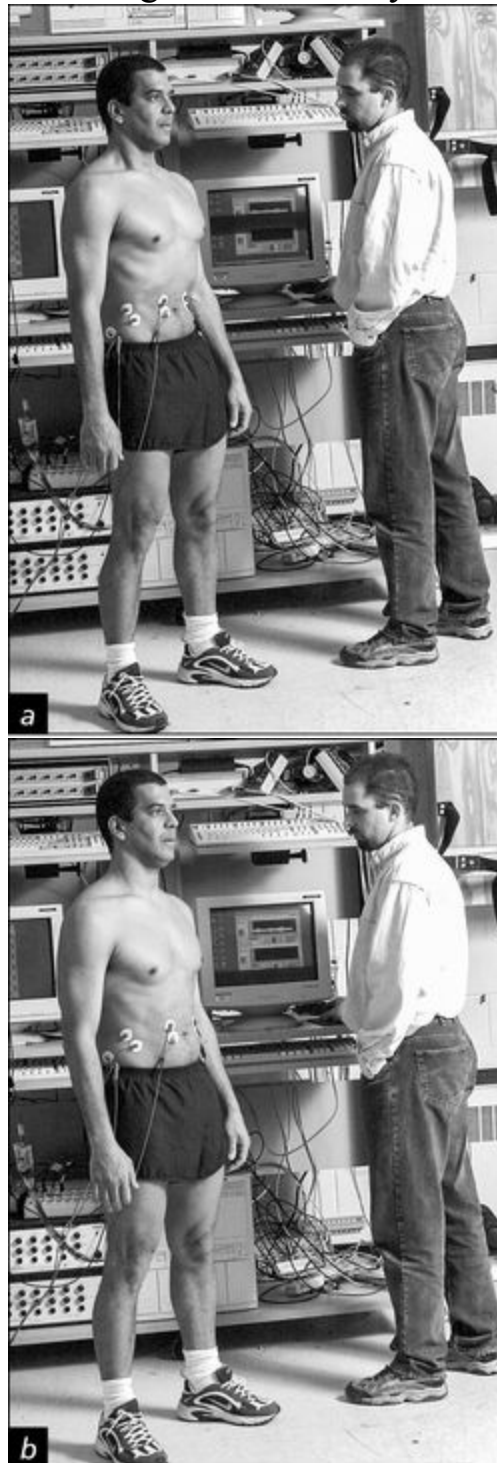


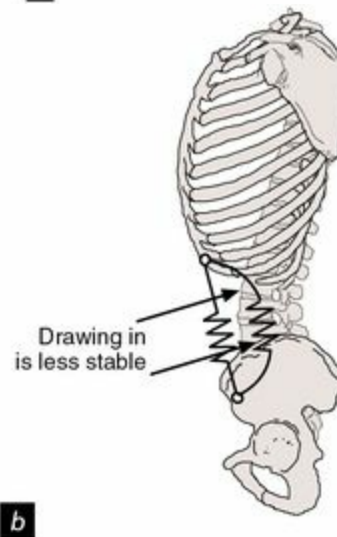
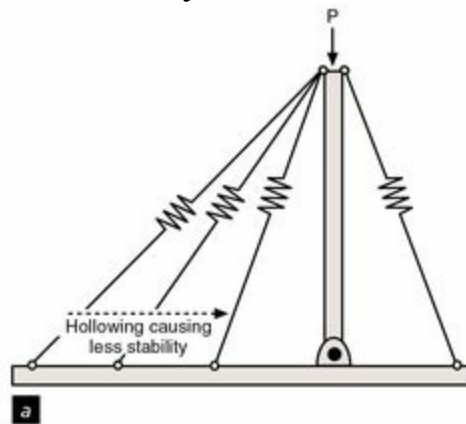
Figure 8.9 (a) Hollowing involves the sucking in of the abdomen to activate transverse abdominis. (b) On the other hand, the abdominal brace activates the three layers of the abdominal wall (external oblique, internal oblique, transverse abdominis), with no drawing in. The cue to the patient is to push the fingers out laterally.



Photos from Stuart McGill

Two mechanisms can shed some light on this issue of hollowing versus bracing. First, the supporting “guy wires” are more effective when they have a wider base (see [figure 8.10](#), a-c); that is, when the abdomen is not hollowed. Second, the obliques must be active to provide stiffness with crisscrossing struts, which measurably enhances stability. Bracing enhances oblique activity, whereas hollowing inhibits it. The lumbar torso must prepare to withstand all manner of possible loads, including steady-state loading (which may be a complex combination of flexion–extension, lateral bend, and axial twisting moments); sudden, unexpected complex loads; and loads that develop from planned ballistic motion. Stiffness is required in every rotation and translation axis to eliminate the possibility of unstable behavior. The abdominal brace ensures sufficient stability using the oblique cross-bracing, although high levels of cocontraction are rarely required—usually about 5 to 10% of maximal voluntary contraction (MVC) levels of the abdominal wall during daily activities are sufficient. The patient must match the level of contraction to the needed stability—there is no need to crush the spine with overcontraction.

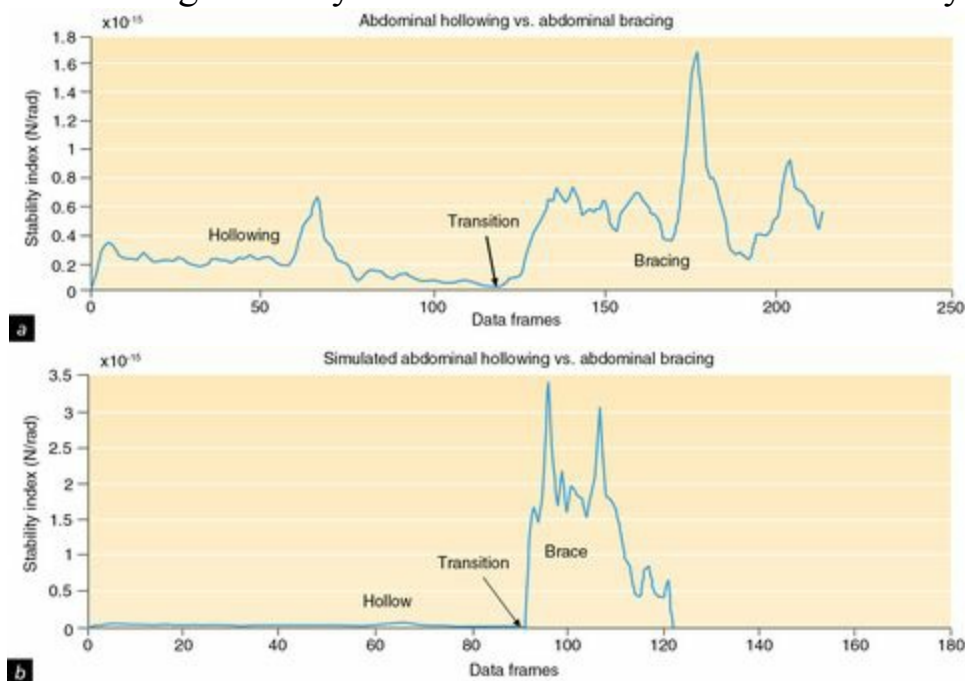
Figure 8.10 (a) Hollowing the muscles reduces the size of the base of the guy wires, as well as the incidence angle at which they attach to the spine. (b) This inherently reduces their contribution to spine stiffness in various modes, which compromises spine stability. (c) Bracing assists in keeping a wide base to the guy wires and recruits the oblique muscle to supply cross-bracing struts for stability in all axes.





A quantitative comparison of the hollow and the brace is clearly seen in a person standing upright with loads in the hands. Simply hollowing can cause the stability index to drop to low levels or even negative levels, which indicates the possibility of instability (see [figure 8.11](#)). Bracing increases the positive stability index value. The subject in these figures is typical in that even though the hollow was taught to target the transverse abdominis, all abdominal muscles were activated when measured. Thus, stability was created while attempting hollowing, although true bracing is superior to create lumbar stability. If a true hollow is accomplished with just the transverse abdominis, as simulated in [figure 8.10b](#), stability is low compared to that with a brace in which the three layers of the abdominal wall are activated ([figure 8.10c](#)). Simply hollowing causes the stability index to drop to negative levels when the load is placed in the hands, compressing the spine. A negative value indicates that instability is possible. In contrast, bracing maintains a positive stability index value, eliminating the possibility of buckling.

Figure 8.11 (a) An example comparing the hollow with the brace (higher stability) in a person standing upright, arms at the sides, with loads placed in the hands. The problem for patients is that isolating transverse abdominis is virtually impossible. (b) A “perfect” hollow was created with simulation and shown to be significantly inferior to the brace to create stability.



Teach Abdominal Bracing

Generally, to demonstrate abdominal bracing to the patient, we stiffen one of our own joints, such as an elbow, by simultaneously activating the flexors and extensors. The patient then palpates the joint both before and after we stiffen it. Then we ask the patient to attempt to stiffen her own joint through simultaneous activation of flexors and extensors. Once she can stiffen various peripheral joints, we demonstrate (again on ourselves, with patient palpation) the same technique in the torso, achieving abdominal bracing. Finally, we again ask her to replicate the technique in her own torso (see [figure 8.12](#)). Occasionally, we use a portable electromyographic (EMG) monitor so the patient can learn through biofeedback what, for example, 5%, 10%, or 80% of maximal contraction feels like (see [figure 8.13](#)). We use similar devices to teach patients how to maintain the contraction while on a wobble board and in functional situations such as when picking up a child,

getting on and off the toilet, and getting in and out of cars.

Figure 8.12 (a) A good cue is for the clinician to place the fingers into the lateral obliques, or have the patient do this themselves. (b) The instruction is to stiffen and push the fingers out laterally.

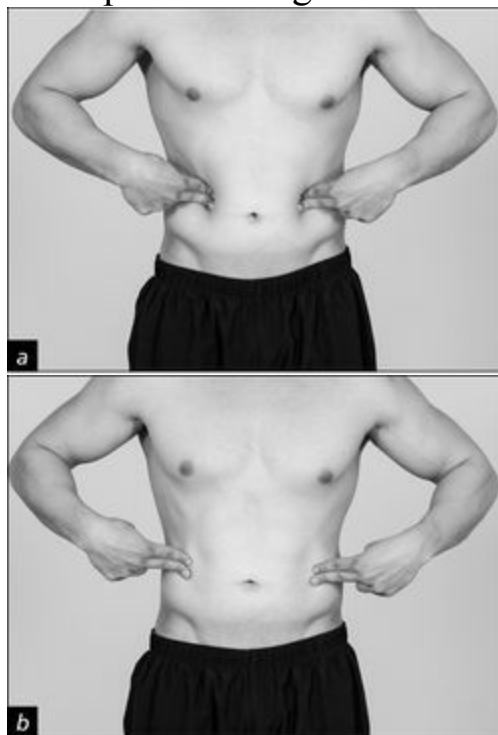


Figure 8.13 EMG biofeedback devices are an economical way to provide feedback to the patient regarding the level of abdominal activation during any type of functional task, from standing on a wobble board to getting into a car.



Photos from Stuart McGill

Given our research on the importance of spine position awareness to spare the spine and our experience in teaching positioning, we became interested in proprioceptive training for the back. The fact that very little evidence was available to validate the use of proprioceptive rehabilitation for the lumbar spine motivated our recent work on spine proprioception (Preuss, Grenier, and McGill, 2005). The purpose of this work was to quantify the effects of a 6-week rehabilitation program to improve lumbar spine position sense and sitting balance. Twelve subjects with a previous history of low back pain were evenly split into a training group and a control group. The subjects in the control group received no intervention, whereas the subjects in the training group received a 20-minute rehabilitation session three times per week emphasizing spine stabilization exercises with a neutral spine. The ability to correct lumbar spine position during four-point kneeling and while sitting substantially improved in the training group over the study. This small but initial study demonstrated that proprioception and position awareness in the lumbar spine can improve through active rehabilitation.

Mental Imagery

The use of mental imagery is helpful for both spine position and muscle activation awareness. Following is a general protocol that we have adapted from the imagery literature for use with spine training.

1. Focus on feeling the surface under the feet or buttocks. Whatever body part is touching a surface, be aware of the sensation.
2. Practice simple motions such as tightening and then relaxing specific muscles in different areas of the body. Then graduate to performing the abdominal brace.
3. Palpate, and have the patient self-palpate, the muscle involved while he is attempting to tighten and relax it. Sometimes a full-body mirror is helpful. The focus for the patient is on the specific muscle(s) involved.
4. Perform motions slowly, chunking them into segments and sequences; then visualize the total motion. For example, beginning with a simple task such as a forward reach, visualize the neutral spine, then activating the extensors and the bracing abdominal muscles, and finally the motion about the hips.
5. Practice the imagery independent of physical action. Of course, the patient will have already been successful in learning spine position awareness, proper muscle control, and desirable motion patterns.

Source: Kathryn McGill, sports psychology consultant

Build Squat Patterns

To have a healthy back, a person needs good gluteal muscle function, and function demands balanced hip power about each axis. This section describes some hip motor patterns that inhibit spine-sparing patterns, and documents several training progressions to address them.

The crossed-pelvis syndrome is a term given to the condition in which the gluteal complex appears to be inhibited during squatting patterns; this syndrome is very common in those with a history of back troubles (and in those who sit for extended periods during work). Interestingly, we still do not know whether the crossed-pelvis syndrome exists prior to back troubles or is

a consequence of having them. Nonetheless, the syndrome is noticeable in many patients referred to our research clinic. This results in two concerns. First, those with aberrant gluteal patterns cannot spare their backs during squatting patterns because they use the hamstrings and erector spinae to drive the extension motion. Subsequently, the erector spinae imposes unnecessary loads on the lumbar spine. In this way, healthy gluteal patterns are needed to spare the back. Second, it is impossible to rebuild optimal squat performance, either for strength or hip extensor power, without well-integrated hip extensor patterns. In fact, the failure of many people to properly rehabilitate is due to neglect of the ability to squat and rise off the toilet, or egress from a car, or walk up stairs, as a result of aberrant gluteal patterns that have not been addressed.

Retraining the gluteals cannot be performed with traditional squat exercises that use a machine. Performing a traditional squat requires little hip abduction. Consequently, there is little gluteus medius activation, and the gluteus maximus activation is delayed during the squat until lower squat angles are reached. This is well documented in McGill (2014). In contrast to the traditional squat, a one-legged squat activates the gluteus medius immediately to assist in the frontal plane hip drive necessary for spine-sparing function together with sooner integration of gluteus maximus during the squat descent motion.

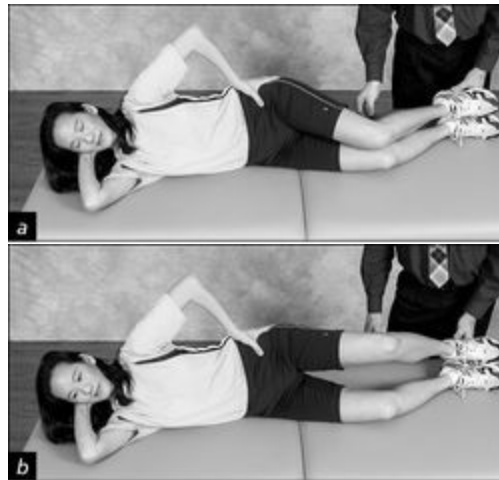
Learning to Activate the Gluteals

The following maneuvers will enable most patients to learn how to isolate and activate both gluteus medius and gluteus maximus.

Gluteus Medius

The first stage involves isolating gluteus medius. Once again, the patient needs to feel the muscle and perceive its activation. The patient lies on her side, places her thumb on the anterior superior iliac spine (ASIS), and reaches with the fingers posteriorly—the tips will be over the gluteus medius (a). With the hips and knees flexed, the patient spreads the knees apart like a clamshell, with the feet remaining together and acting as a hinge (b). The patient feels with the fingers the activation of gluteus medius. This maneuver is to simply activate the gluteus medius and should not be considered a

strengthening exercise. There is no need to offer resistance at this stage (resistance is imposed later during strength training). True isolation of the gluteus medius is not possible; other muscles are active. In this posture, the external hip rotators are recruited. Recent work of Sidorkewicz, Cambridge, and McGill (2013) has shown that although muscles such as the tensor fascia latae are activated, the gluteus medius dominates this exercise, and the changing hip flexion does not substantially influence the relative activation of these two muscle groups.



Gluteus Maximus

Lying on the back with the knees flexed and the feet on the floor, the patient places the fingers on gluteus maximus to feel its activity. Have the patient imagine a coin placed in the gluteal fold that must not be dropped. The patient activates gluteus maximus by squeezing the buttocks—not by creating hip extension. The focus is on the pelvis at this stage to ensure that no pelvic tilting occurs. The lumbar spine remains in neutral posture (a). Then, once the activation has been mastered, the patient begins bridging the torso off the floor. The clinician at this stage palpates the hamstrings (b). Those who are hamstring dominant and gluteal deficient will immediately activate the hamstrings just prior to the occurrence of motion. This pattern is very dominant in those who have the aberrant crossed-pelvic syndrome, but is also seen in some sport-specific athletes such as cyclists. The patient must repeatedly try to begin the bridging action without hamstring activity (or at least only mild activity).



For some patients, overriding the hamstring-dominant tendency requires coaching and cuing from the clinician. For these challenging cases, we place our foot against the patient's toes and instruct the patient to continue with the preparatory gluteal activation but then also very slightly activate the quadriceps by very mildly attempting to extend the knees (c). Buttrressing the patient's feet with the clinician's foot assists this. A gentle stroke on the quads to assist the patient's imagining and perception of mild knee extension also facilitates this pattern to enhance gluteal dominance. Then the patient repeats the attempt to bridge with gluteal dominance. Once this is mastered, squat performance will improve.

Imagining squeezing the gluteus maximus muscles prior to performing

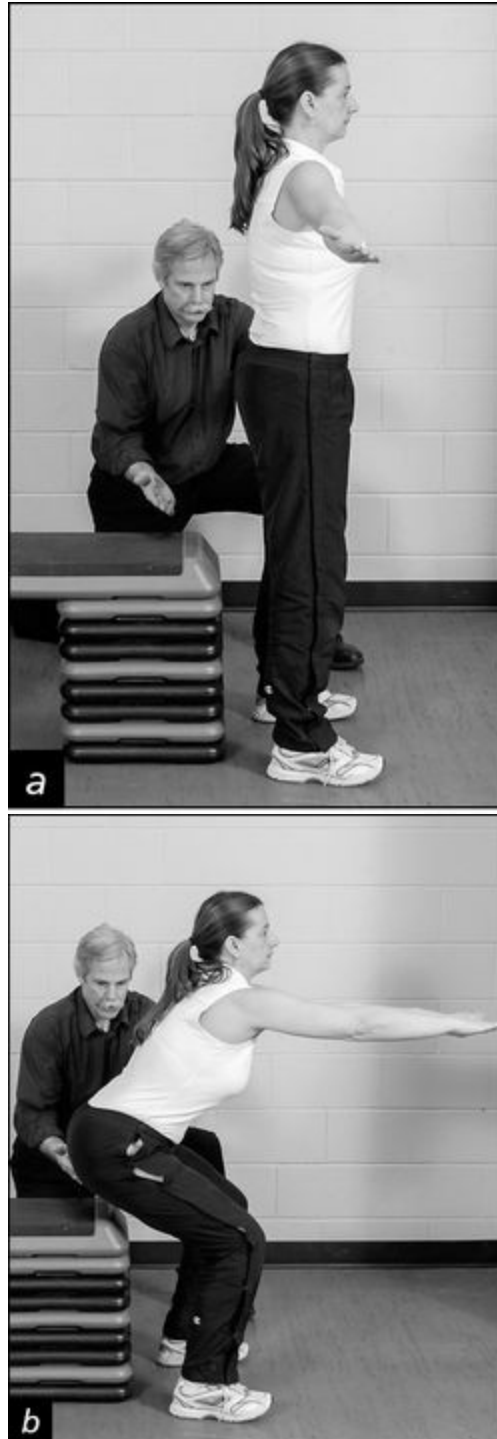
the back bridge will assist in grooving gluteal-dominant hip extension patterns. Then the patient performs the bridge with a focus on gluteal contraction throughout the full range. For those having difficulty, quadriceps stroking can assist with mild knee extension to further reduce hamstring contribution.

Beginning Basic Squat Patterns

First a word of caution: Do not start this too early with patients who either are disabled or have very painful backs.

Potty Squat

When appropriate, we begin a basic squat progression with a potty squat. Sitting on the corner of a chair or a stool, the patient positions the feet under the body to rise off the chair without using any momentum shifts. The lumbar spine is neutral and braced. This begins to groove a good two-legged squat position. Then, with progression to a standing position, the arms are held out laterally (a) and moved in front of the body as the patient squats (b). Of course, emphasis is on maintaining a neutral lumbar spine and abdominal bracing. The hips follow a trajectory along a line about 45° from the vertical. “Squat back” is a better instruction than “Squat down.” The motion is predominantly at the hips, which is known as the hip hinge.



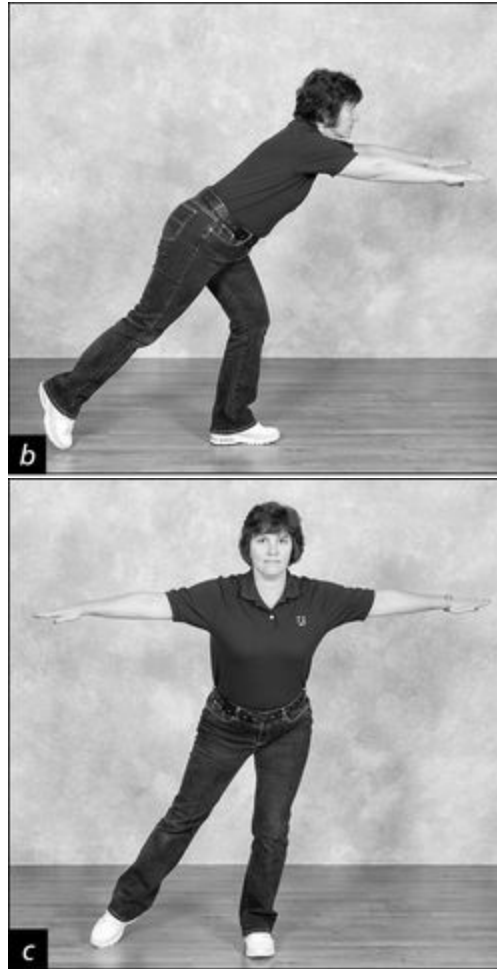
Photos from Stuart McGill

Single-Leg Squat

A single-leg squat involves the same arm motion as the potty squat to assist balance. As the single-leg squat is performed, the free leg is held

behind and the knee is touched to the floor, or the toe is reached with an outstretched leg to a distant object behind. (a , b) Then, the free leg is reached out to a distant object placed laterally during the squat (c). Variations include working the free leg to various positions around the clock. This challenges the full hip extensor, flexor, and abduction torque generators together with keen motor control. Full integration with the pelvis and lumbar spine is achieved with emphasis on the appropriate motor and motion patterns. Specific focus is on maintaining a neutral lumbar spine. Focus on hip motion, the gluteal muscles producing the hip extension torque, and a stiff torso. Caution with anterior foot placement is necessary with some patients who are unable to maintain a neutral spine.



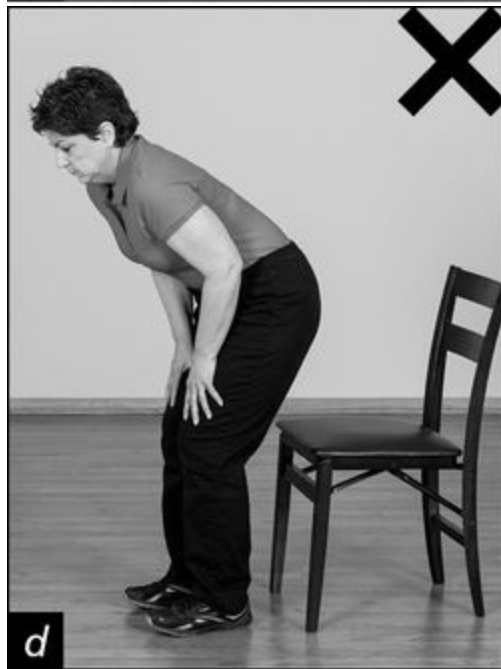


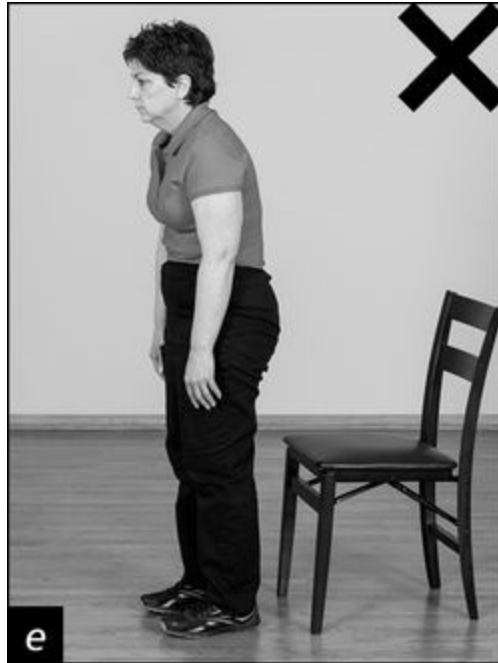
The single-leg squat follows the same hip-hinging motion as the potty squat. The nonsupport leg is projected to the rear and to the side. Be cautious with forward projection of the foot to the front, because it often causes spine flexion. The abdominal muscles are braced, the lumbar spine is in neutral, and the mental focus of the patient is on hip extension torque. Notice that the hips are drifted posteriorly during the descent to place more emphasis on the gluteals for hip extension, unloading both the knees and the back.

Rising From a Chair

Rising from a chair follows the potty squat mechanics. Faulty motion patterns include initiating the motion with spine flexion. Instead, the spine should be extended with the rib cage rising and the hips flexed. The feet are drawn underneath the patient, the knees and feet are spread, and the hips externally rotate to integrate the gluteal muscles for hip extension.







(a, b) The first faulty motion is initiated with spine flexion. (c) Then the hips are extended with the hamstrings, and (d) the spine is extended with the back extensors. Poor technique is characterized by spinal flexion and lifting from the back. (e) The stooped standing posture results from a failure to pull the hips through with extension.







Corrected rising from the chair begins by spreading the knees and placing the feet under the client's center of mass (a). Then the torso leans forward with hip flexion, not spine flexion (b). Sniffing the air first helps with form. Then the hips are pulled through as the hands slide up the thighs (c), until an upright posture is achieved (d).

Stage 2: Establish Stability Through Exercises and Education

For patients to achieve stability not just during rehab, but also in all aspects of their lives both during and after rehab, both exercise and education are crucial. The fundamental tenet of human motion is that a proximal joint must be stiffened and stabilized before a distal joint can produce force, torque, and movement. Skilled stiffness and stability techniques must be established.

- Establish stabilizing motor patterns through stabilization exercises. Generally, stage 2 should begin with stabilization exercises. The trick is

to find an appropriate starting level. For those who arrive at our university clinic having failed traditional therapy, we generally undershoot what many would consider an appropriate loading level. The typical progressive improvement philosophy of work hardening with weekly improvement goals does not work for these people; they require more patience. They will have good and bad days, but the general positive slope in improvement must be established—however gradual. Once we can document a positive slope in improvement, then we can increase the rehabilitation challenge. Specific exercises are discussed and illustrated in chapter 10.

- Ensure stabilizing motion patterns and muscle activation patterns during all activities. You must clarify the range of activities (e.g., daily living, occupational, athletic) for which the patient must be prepared. This is obtained through interview. Although there is no standard form, we begin by documenting the patient's daily routine, which includes occupational demands and those of daily living. Previous chapters offered many examples of sparing the spine with an appropriate posture and muscle activation pattern. The guidelines in the next section will help you determine what exercises to prescribe for patients in this stage. In addition, we spend time rehearsing daily activities to be sure the patient is learning and using spine-sparing motion and motor (muscle activation) patterns.

Stage 3: Develop Endurance

Endurance is necessary for maintaining stabilizing patterns of muscle activity. There is a progression to building endurance, and it starts with building endurance without becoming tired! Here is what you should keep in mind as you progress patients toward greater endurance:

An Important Reminder

Remember, because half the battle is to remove the pain triggers, make sure to follow the recommendations of the previous chapters.

- Typically, initial endurance is built with repeated sets of relatively short holds. Holds should be no longer than 7 or 8 seconds. The duration is based on recent evidence from near-infrared spectroscopy that indicates a rapid loss of available oxygen in torso muscles contracting at these levels. A short relaxation of the muscle restores oxygen (McGill, Hughson, and Parks, 2000). The endurance objectives are achieved by increasing the repetitions of the exertions rather than the duration of each hold.
- Identify endurance deficits. Motivated by the evidence of the superiority of extensor endurance over strength as a benchmark for good back health, we recently documented normal ratios of endurance times for the torso flexors relative to the extensors and lateral musculature (see chapter 9). Use these values to identify endurance deficits—both absolute values and values for one muscle group relative to another.
- Use the reverse pyramid for endurance training. This approach to designing endurance sets is founded in the Russian tradition of maintaining excellent technique and form. The idea is to train endurance without becoming tired. For example, a training session of sets for the side bridge exercise using five repetitions would look like this:
 1. Five repetitions on the right side
 2. Five repetitions on the left side
 3. Rest a few seconds
 4. Four repetitions on the right side
 5. Four repetitions on the left side
 6. Rest a few seconds
 7. Three repetitions on the right side
 8. Three repetitions on the left side
 9. Finished

Good technique is facilitated as the repetitions are reduced with each fatiguing set. This is generally done to build the endurance base, because the objective of maintaining sufficient oxygen levels is met so that failure is not a result of oxygen starvation or acidic metabolite buildup. This is the approach for those with back pain. Endurance training progressions involve much longer durations for more athletic patients and are described in McGill (2014).

Summary: Checklist for Patient Progression

Clinical Relevance

As we have noted, the first three stages of our five-stage program focus on rehabilitation. Following is an expanded checklist for rehabilitation that adds some critical elements.

1. Identify and remove the exacerbating activities.
2. Have the patient record in a journal daily how the back feels as well as the tasks and activities performed.
3. Identify and correct perturbed motion and motor patterns using corrective exercise. Develop spine position awareness (hips vs. lumbar motion) and the ability to maintain appropriate levels of abdominal bracing. Establish spine-sparing motion and motor patterns.
4. Begin appropriate spine exercise and appropriate stabilization and mobilization tasks.
5. Develop muscular endurance.
6. Transfer to daily activities.

Note that training athletes in our program involves the same stages with the addition of two—training strength and power (see chapter 11 and McGill, 2014).

Guidelines for Developing the Best Exercise Regimen

Reports of the effectiveness of training and rehabilitation programs for the low back are quite variable: some claim great success, whereas others report no success or even negative results (Faas, 1996; Koes et al., 1991). The discrepancy regarding the effectiveness and safety of exercise programs is probably due to clinicians prescribing inappropriate exercises because they do not understand the tissue loading that results during various tasks. The approach described in this book has been used with great success for preventing back disorders (e.g., the Finnish military; Suni et al., 2006). Resist the urge to enhance mobility just to adhere with the disability rating system. That system is for legislative convenience only. Rather, judge your success by how well you are able to reduce patients' pain and restore their ability to complete tasks. Also, there must be a reason to perform an exercise, and to perform it in a specific way. Chapter 9 will help you identify deficits in patients so that you can address them in a specific program. If there is no reason to prescribe an exercise, then don't—work only within the restricted capacity of the patient to achieve specific goals.

Developing a Sound Basis for Exercise Prescription

I have quantified and selected the exercises in the following chapters based on tissue-loading evidence and the knowledge of how injury occurs to specific tissues (described in previous chapters). Choosing exercises, however, still involves making the best educated guess, which is developed through clinical experience. The following example illustrates the need for quantitative analysis to evaluate the safety of certain exercises.

We have all been told to perform sit-ups and other flexion exercises with the knees flexed—but on what evidence? Several hypotheses have suggested that this disables the psoas or changes the line of action of the psoas, or both. Magnetic resonance imaging (MRI)-based data (Santaguida and McGill, 1995) demonstrated that the psoas line of action does not change as a result of lumbar or hip posture (except at L5-S1) because the psoas laminae attach to each vertebra and follow the changing orientation of the spine. However, there is no doubt that the psoas is shortened with the flexed hip, modulating

force production. But the question of whether there is a reduction in spine load with the legs bent remains. In 1995 McGill examined 12 young men using the laboratory technique described previously and observed no major difference in lumbar load as the result of bending the knees (average moment of 65 Nm in both straight and bent knees; compression of 3,230 N with straight legs and 3,410 N with bent knees; shear of 260 N with straight legs and 300 N with bent knees). Compressive loads in excess of 3,000 N certainly raise questions of safety. This type of quantitative analysis is necessary to demonstrate that the question of whether to perform sit-ups using bent knees or straight legs is probably not as important as the question of whether to prescribe sit-ups at all! There are better ways to challenge the abdominal muscles.

Basic Issues in Low Back Exercise Prescription

Several exercises are required to train all the muscles of the lumbar torso, but which exercises are best for a given person? This determination depends on a number of variables, such as the person's fitness level, training goals, history of previous spinal injury, and other factors. However, depending on the purpose of the exercise program, several principles apply. For example, someone beginning a postinjury program would be advised to avoid loading the spine throughout the range of motion, whereas a trained athlete may indeed achieve higher performance levels by doing so. Another general rule of thumb is to preserve the normal low back curve (similar to that of upright standing) or some variation of this posture that minimizes pain. Although in the past many clinicians have recommended performing a pelvic tilt when exercising, this is not justified; we now know that the pelvic tilt increases spine tissue loading, because the spine is no longer in static-elastic equilibrium. Thus, the pelvic tilt appears to be contraindicated when challenging the spine. Basic issues you should consider when prescribing exercises for low back rehabilitation are discussed in the following sections.

Flexibility

The decision of whether to train for optimization of spine flexibility depends on the person's injury history and exercise goal. Generally, for the

injured back, spine flexibility should not be emphasized until the spine has stabilized and has undergone endurance and strength conditioning—and some may never reach this stage! Despite the notion held by some, few quantitative data support the idea that a major emphasis on trunk flexibility improves back health and lessens the risk of injury. In fact, some exercise programs that have included loading of the torso throughout the range of motion (in flexion–extension, lateral bend, or axial twist) have had negative results (e.g., Nachemson, 1992), and greater spine mobility has been associated with low back trouble in some cases (e.g., Biering-Sorensen, 1984). Further, research has shown that spine flexibility has little predictive value for low back trouble (e.g., Sullivan, Shaof, and Riddle, 2000).

In the context of trying to stretch the back and train flexibility, the insightful work of Solomonow and colleagues (2002) has shown that the stretch reflex is diminished and muscle spasms can result. The most successful programs appear to emphasize trunk stabilization through exercise with a neutral spine (e.g., Hides, Jull, and Richardson, 2001; Koumantakis, Watson, and Oldham, 2005; Saal and Saal, 1989) while stressing mobility at the hips and knees. (Bridger, Orkin, and Henneberg, 1992, demonstrated advantages for sitting and standing, and McGill and Norman, 1992, outlined advantages for lifting.) Note that stretching programs that have enhanced hip range of motion used several static and dynamic approaches and considered technique directed to muscle, nerve, fascia, and the joint connective tissues in all directions (Moreside and McGill, 2012).

Finally, removing lumbar flexion from morning activities substantially improves patients, on average (Snook et al., 1998). Despite this evidence, many patients are still instructed to pull their knees to their chest in the morning and perform toe touches (see [figure 8.14](#), a-c). The destabilizing consequences of full flexion were described in chapters 3 and 4.

Figure 8.14 “Silly stretches,” such as (a, b) pulling the knees to the chest and (c) toe touches, are often prescribed to patients to do in the morning. These can cause instability! Paradoxically, the stretch receptors in the back are stimulated, providing a false sense of relief that may last about 20 minutes. In reality, they increase pain sensitivity later.



Photos from Stuart McGill

On occasion, a patient will want to mobilize the spine. There are also therapeutic reasons for mobilizing the spine through a range—for example, gentle movement postsurgery to prevent adhesions and short scarring of

neural and connective tissues. We have quantified that the most spine- and disc-sparing method is to adopt the hands and knees position and perform the cat–camel exercise (see chapter 10). This unloaded range of motion cannot be achieved in a standing posture and is suited to those concerned with safety. Those interested in specific athletic activities may sometimes be an exception to this rule. (Of course, spine flexibility may be more desirable in athletes who have never suffered back injury.)

Strength

In general, strength seems to have little to do with back health even though increasing torso muscle strength is a popular objective of low back rehabilitation protocols (note that the issue is rehabilitation and not athletic performance). This is not to imply that strength is not important. Rather, it is to emphasize that the way the spine moves is relatively more important in terms of spine health.

Back muscle strength in particular has not been found to be a significant predictor of first-time injury. (In the context of cause and effect, predicting first-time injury offers special insight.) Only Troup, Martin, and Lloyd (1981) found, while testing torso muscles, that reduced dynamic strength was a predictor of recurring back pain. However, in a prospective study, Leino, Aro, and Hasan (1987) found that neither isometric nor dynamic trunk strength predicted the development of low back troubles over a 10-year follow-up period. The Biering-Sorensen (1984) study, previously noted, found that isometric back strength did not predict the appearance of low back trouble in previously healthy subjects over a 1-year follow-up. Holmstrom and Moritz (1992) recorded reduced isometric trunk extensor endurance times in male workers with low back disorders (LBDs) compared to those without, but found no differences in isometric flexion or extension strength.

Strength appears to have little, or a very weak, relationship with low back health. Strength is for athletic performance objectives. In contrast, muscle endurance, when separated from strength, appears to be linked with better back health.

Our data have suggested that although having a history of low back troubles is not related to reduced strength, it is related to a perturbed flexion-to-extension strength ratio (McGill et al., 2003). This difference in the ratio appeared to be mainly influenced by greater extensor strength relative to

flexor strength in those with disorders. However, the most influential ratio was strength to endurance (see the next section for an explanation).

Endurance

Two cross-sectional studies, those of Nicolaisen and Jorgensen (1985) and Alaranta and colleagues (1994), found reduced extensor endurance in workers who reported disabling low back pain. Both Biering-Sorensen (1984) and Luoto and colleagues (1995) suggested that although isometric strength was not associated with the onset of back troubles, poor static back endurance scores were. Some have expressed concern that patients with poor muscle endurance scores have poor scores from a lack of effort (a psychological variable) rather than any physiological limitation. A study by Mannion and colleagues (2001) suggested that of the total variance measured in the endurance scores of their back patients, 40% was explained by physiological fatigue (quantified as declines in the EMG power spectrum), and only 10% was explained by psychological variables (quantified as motivation and fear of pain variables, from a questionnaire). Another study (McGill et al., 2003) suggested that having a history of low back troubles appears to be associated with a different flexion-to-extension endurance ratio—the extensors have less endurance and the flexors have more endurance. This imbalance in endurance also appeared between the right and left lateral musculature as evidenced by the asymmetry in right and left endurance holding times (e.g., right side bridge/left side bridge (RSB/LSB) ratio of 0.93 for those with an LBD history vs. 1.05 for those without). The next issue addressed the question of whether strength and endurance are related. Interestingly, the flexor strength (Nm)-to-endurance (sec) ratio was between 3 and 3.5 for the flexors in both those with normal backs and those with a history of repeated episodes of disabling pain, and for the extensors of those with normal backs. The ratio was much larger for the extensors (5.3, $p = .033$) in those with a history of LBD.

Our most recent work with firefighter and police populations simply enhanced our historical impressions. It appears that many get hurt in the training room when focusing on strength training. The stronger men were not protected from back injury. We now believe that the key is to develop skills to control movement with the available strength. People get injured when they break form. More strength will not prevent this. Rather, enhanced endurance coupled with control skills enables people to maintain good form

during repeated strength exertions. This is why endurance must be established before strength is enhanced in those recovering from repeated and recurrent back pain episodes.

In summary, several studies have suggested that diminished trunk extensor endurance and not strength is linked to low back troubles. Recent data confirm this notion and enhance it further by suggesting that the balance of endurance between flexor and extensor muscles and the balance between the right and left sides of the torso appear to be linked to a history of back troubles severe enough to result in work loss. Note that these were losses that lingered long after the disabling episode. In those with a history of disabling LBD, the average length of time since the last work loss episode was 261 weeks (standard deviation = 275), whereas the average length of time lost from work in that episode was 7 days (standard deviation = 10). The lesson for exercise prescription is that graduated, progressive exercise programs (i.e., of longer duration and lower effort) that emphasize endurance before strength development are preferable.

Aerobic Exercise

The role of aerobic exercise in both reducing the incidence of low back injury (Cady et al., 1979) and treating low back patients (Juker et al., 1998) is compelling. Much new knowledge has already been introduced regarding the analgesic effects of aerobic exercise at the molecular level. An investigation into loads sustained by the low back tissues during walking (Nutter, 1988) confirms very low levels of supporting passive tissue load coupled with mild, but prolonged, activation of the supporting musculature. Callaghan, Patla, and McGill (1999) documented that fast walking with the arms swinging results in lower oscillating spine loads. When tolerable, aerobic exercise, particularly fast walking, appears to enhance the effects of back-specific exercise.

Many patients reveal torso and pelvis frontal plane strength and endurance deficits. Our recent work quantifying walking and carrying loads with one and two hands showed this to be an excellent progression for addressing this deficit. The approach is tolerable to most pained patients. Fast-paced walking creates the aerobic stimulus and the frontal plane challenge necessary to train the musculature to support the pelvis in a stiffened position. This facilitates stance and leg swing with no pain. We start

some patients with simple weight transfer from side to side, then progress to marching in place, then walking (see [figure 8.15](#)). Carrying continues the progression, recognizing that load in just one hand results in more compressive force on the spine than the same load in each hand (McGill, Marshall, and Andersen, 2012).

Figure 8.15 (a) Walking progressions begin with weight transfer from one leg to the other, followed by (b) learning to control forward lean using the ankles, called the leaning tower. Move on to marching and then walking. The focus should be on spine and pelvic stiffness and control, while freeing the hips. (c) Poor form; (d) good form.







Order of Exercises Within a Session

Because the spine has a loading memory, a prior activity can modulate the biomechanics of the spine in a subsequent activity. For example, if a person sat in a slouched posture for a period of time sufficient to cause ligamentous and disc creep, she would have residual ligament laxity for a period of time. (We have measured laxity of over a half hour in some cases; McGill and Brown, 1992.) The nucleus volume appears to redistribute upon adoption of a standing posture (Krag et al., 1987). This redistribution takes time. If the spine is flexed in one maneuver, then it probably should return to neutral or extension for the next.

Viscosity is another property of biological tissues—in this case, a frictional resistance to motion within the spine and torso tissues. This is why motion exercises (such as the cat–camel) are usually performed first as part of a warm-up; once the viscous friction has been reduced, subsequent motion can be accomplished with less stress.

Establishing Movement Patterns

We have found that patients are best served when we establish the preferred movement patterns for spine stability at the beginning of the session. These exercises produce a residual stiffness in the spine for both pain control and improvement of function.

A final consideration is the need to continually reinforce healthy, joint-conserving, and stabilizing motor patterns. Depending on the exercise objectives, we most often begin an exercise or training session with some spine stabilization exercises to reinforce the patterns that will continue over to other exercises in the program. Our recent work has shown that protective stiffness is retained in the torso tissues following the performance of the big three exercises (Lee and McGill, 2015). Understanding spine biomechanics, function, and pain mechanisms guides the optimal ordering of tasks in a training session.

Breathing

Debate continues regarding training for breathing during exertion. Should one exhale or inhale during a particular phase of movement or exertion?

In the rare cases of very heavy lifting or maximal exertions (which would not be part of a rehabilitation program), high levels of intra-abdominal pressure (IAP) are produced by breath holding using the Valsalva maneuver. This elevated IAP, when combined with high levels of abdominal wall cocontraction (bracing), ensures spine stiffness and stability during these extraordinary demands.

Another motivation given for striving to achieve higher IAP is the need to reduce the transmural gradient in the cranium to lessen the risk of blackout or stroke (McGill, Sharratt, and Seguin, 1995). The explanation for this risk reduction is as follows:

- Building IAP is associated with a rise in the central nervous system (CNS) fluid pressure in the spine, which forms an open vessel to the CNS and brain.
- Upon exertion, enormous elevation in blood pressure occurs (documented in weightlifters to be well over 400 mmHg).
- Pressure in the cranial vessels creates a large transmural pressure gradient that is reduced if the CNS fluid pressure is likewise elevated,

reducing the load on the vascular vessels.

Although this explanation is valid, the mechanism should be considered only for extreme weightlifting challenges—not for rehabilitation exercise.

In the design of rehabilitation exercise, a major objective is to establish spine stabilization patterns. An important feature of stable and functional backs is the ability to cocontract the abdominal wall (abdominal brace) independently of any lung ventilation patterns. Good spine stabilizers maintain the critical symmetrical muscle stiffness during any combination of torque demands and breathing patterns (such as when playing a basketball game). Poor stabilizers allow abdominal contraction levels to cycle with breathing at critical moments when stability is needed. Entraining a particular direction in lung air flow to a particular part of an exertion is not helpful for these patients. This would be of little carryover value to other activities; in fact, it would be counterproductive. As the patient progresses to performance objectives and higher loads with speed, the relationship changes. Consider the tennis players serving the ball with an audible grunt. This overdrives the abdominal wall contraction, creating a superstiffness in the core and enabling higher shoulder velocity and a faster serve. This mechanism and technique is explained in McGill (2014).

In summary, patients susceptible to pain should train to breathe freely while maintaining the stabilizing isometric abdominal wall contractions. Entrained breathing comes later as the progression shifts from eliminating pain to enhancing performance.

Time of Day for Exercise

As pointed out in part II, the intervertebral discs are highly hydrated upon rising from bed; the annulus is subjected to much higher stresses during bending under these conditions, and the end plates fail at lower compressive loads as well. Thus, performing spine-bending maneuvers at this time of day is unwise. Yet many manual medicine physicians continue to suggest that patients perform their therapeutic routines first thing in the morning. This appears to be due to convenience and ignorance. Because the discs generally lose 90% of the fluid that they will lose over the course of a day within the first hour after rising from bed, we suggest simply avoiding this period for exercise (that is, bending exercise) for either rehabilitation or performance

training. Although there hasn't been a study on the enhancements obtained during exercise routines as a function of time of day, Snook and colleagues (1998) did discover that the conscious avoidance of forward spine flexion in the morning improved their patients' backs.

Notes for Rehabilitation Exercise Prescription

Clinical Relevance

Exercise professionals face the challenge of designing exercise programs that consider a wide variety of objectives. Consider these guidelines:

- Although some experts believe that exercise sessions should be performed at least three times per week, low back exercises appear to be most beneficial when performed daily (e.g., Mayer et al., 1985).
- The no pain, no gain axiom does not apply when exercising the low back, particularly when applied to weight training. Scientific and clinical wisdom suggests that the opposite is true.
- Research has shown that general exercise programs that combine cardiorespiratory components (such as walking) are more effective in both rehabilitation and injury prevention (e.g., Nutter, 1988).
- Diurnal variation in the fluid level of the intervertebral discs (discs are more hydrated early in the morning after rising from bed) changes the stresses on the disc throughout the day. People should not perform full-range spine motion under load for 1 to 2 hours after rising from bed (e.g., Adams and Dolan, 1995).
- Low back exercises performed for health maintenance need not emphasize strength with high-load, low-repetition tasks. Rather, more repetitions of less demanding exercises enhance endurance and strength. There is no doubt that back injury can occur during seemingly low-level demands (such as picking up a pencil) and that injury from motor control error can occur. Although the chance of motor control errors that result in inappropriate muscle forces appears to increase with fatigue, evidence also indicates that passive tissue loading changes with fatiguing lifting (e.g., Potvin and Norman, 1992). Given that endurance has more protective value than strength (Luoto et al., 1995), strength gains should not get ahead of endurance.

- No set of exercises is ideal for everyone. An appropriate exercise regimen should consider the person's training objectives, be they rehabilitation, reducing the risk of injury, optimizing general health and fitness, or maximizing athletic performance. Although science cannot be used to evaluate the optimal exercises for each situation, the combination of science and clinical experience will result in enhanced low back health.
- Both patients and clinicians should be patient and stick with the program. Increased function and reduced pain may not occur for 3 months (e.g., Manniche et al., 1988).

A Final Note

The five stages described in this chapter have a rationale and a progression based on scientific evidence of mechanisms and efficacy. Not every patient will need each exercise. However, following the stages greatly enhances the chances for success.